

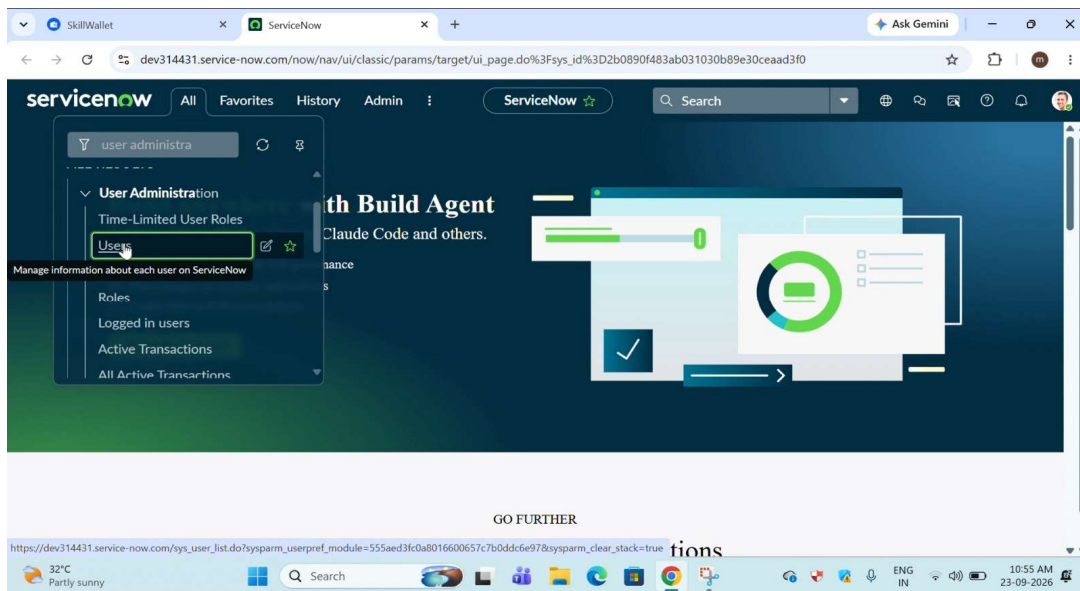
Script-Controlled ACL – Restrict Record Access Based on Field Value

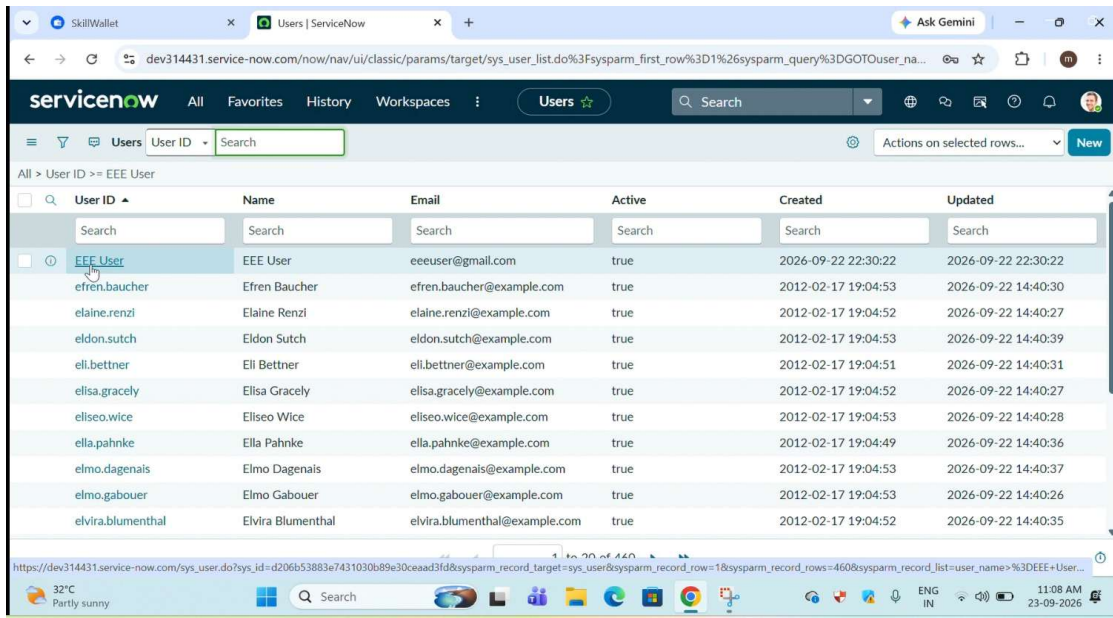
Milestone-1:

Creation of Users and Roles

User Creation:

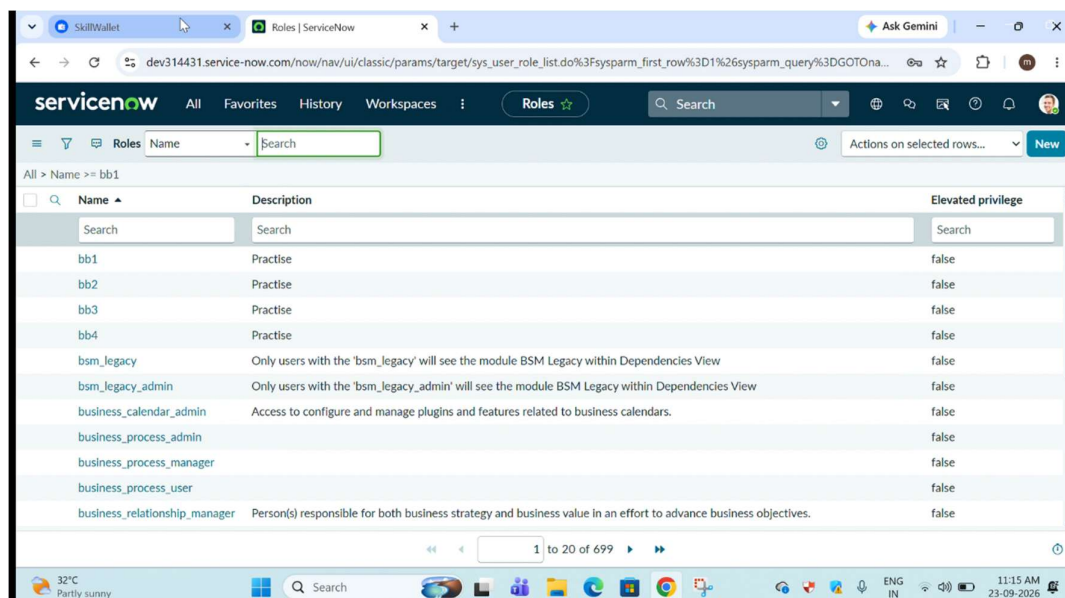
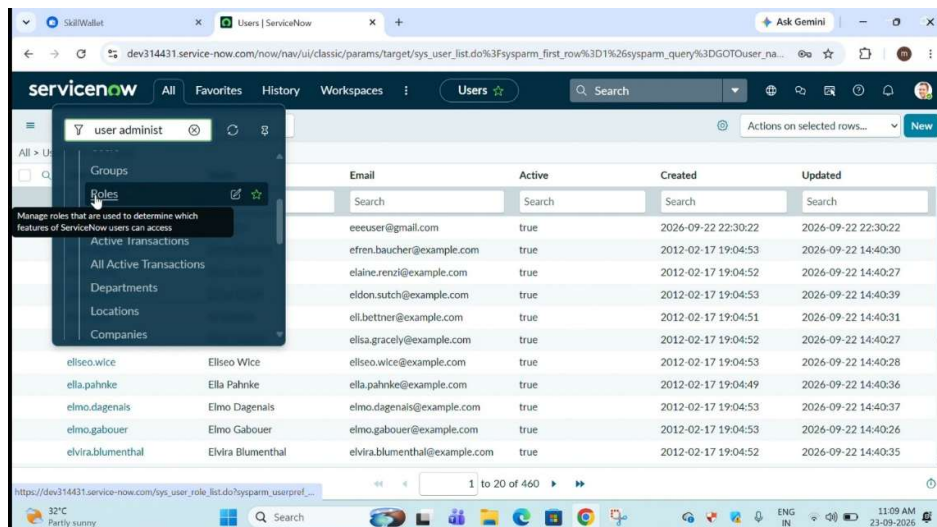
1. **Log in** to the ServiceNow instance with **admin access**.
2. **Create a test user:**
 - Navigate to **User Administration** → **Users**
 - Click **New**
 - **User ID:** EEE User
 - **First name:** EEE
 - **Last name:** User
 - **Email:** eeeuser@gmail.com
 - Save the record





Role Creation

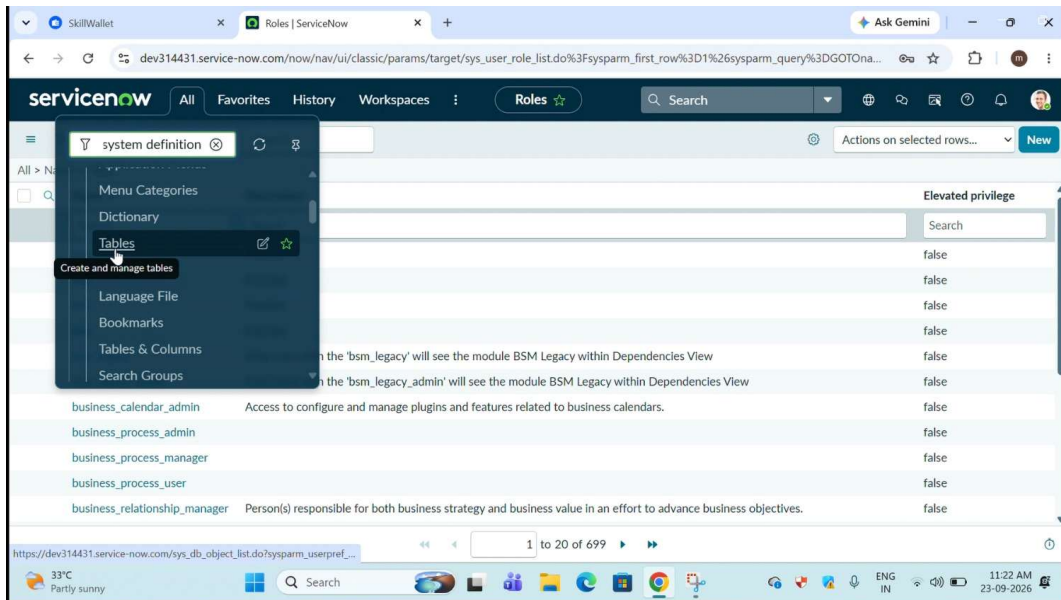
- Navigate to **User Administration** → **Roles**
- Click **New**
- **Name:** Enter role name as **bb1**
- **Description:** Brief role purpose
- Click **Submit**
- Create other roles also like **bb2,bb3,bb4.(follow the same procedure)**
- Assign all the roles to the EEE User



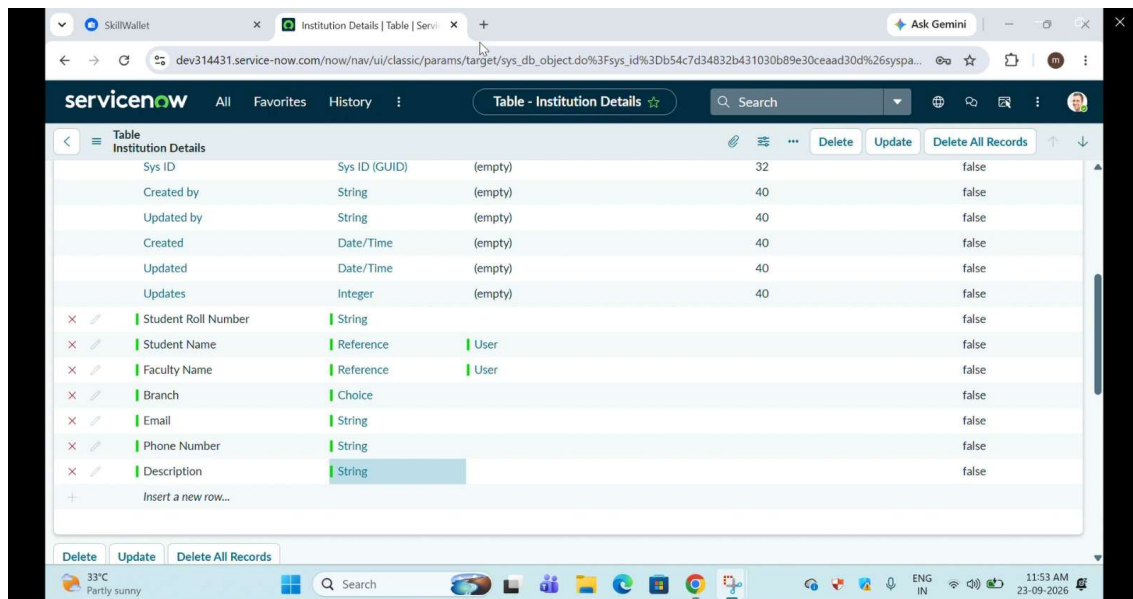
Milestone-2:

Tables Creation:

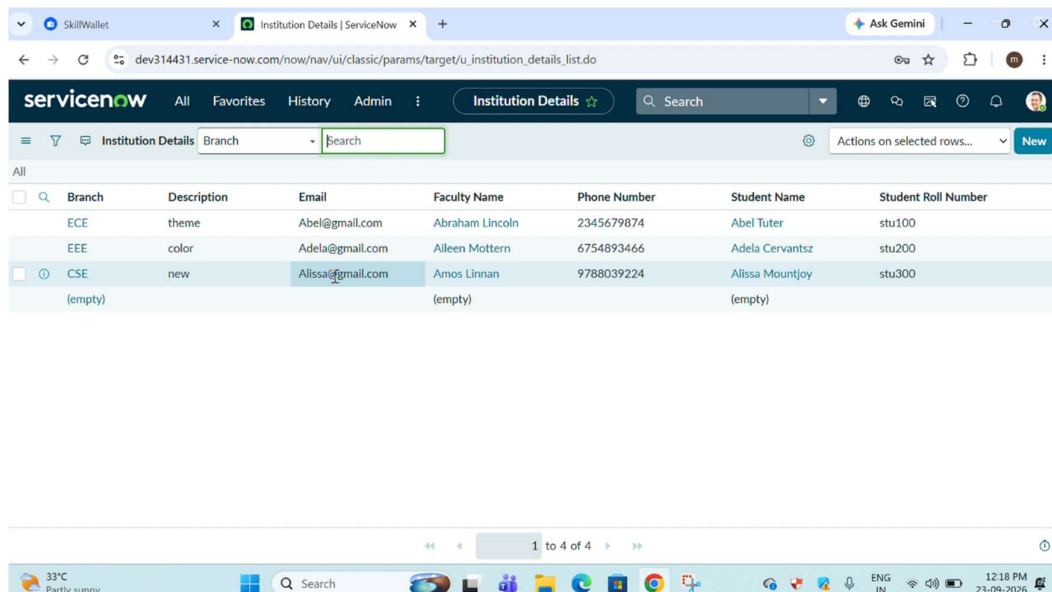
1. In the **Application Navigator**, search for **System Definition** → **Tables**.
2. Click **New** and create a custom table with the following details:
 - **Label:** Institution Details
 - **Name:** u_institution_details
 - **Extends for :** false



3. Click **Submit**.
4. Add the following fields to the **u_institution_details** table:
 - **Student Roll Number**– Auto Number
 - **Student Name** – Reference- User
 - **Faculty Name** – Reference- User
 - **Branch** – Choice (ECE, EEE, CSE)
 - **Email**– String
 - **Phone Number**– String
 - **Description**– Multi String
5. Save the Table



6. Create multiple records in the **Student Records** table with different branch values (ECE, EEE, CSE).



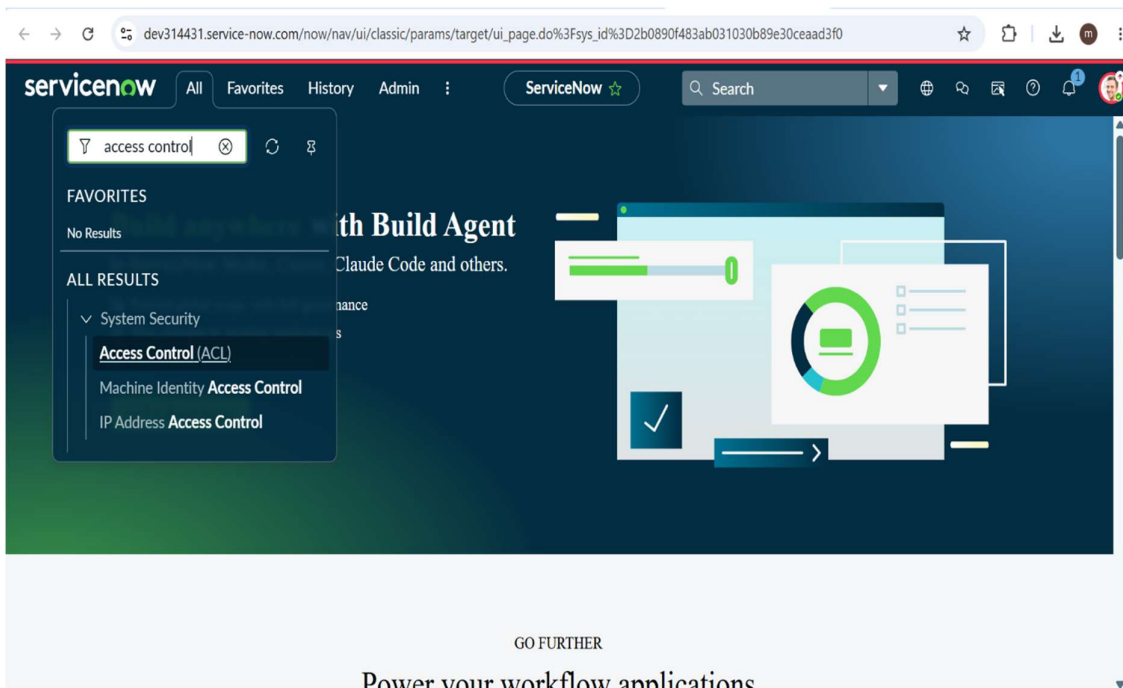
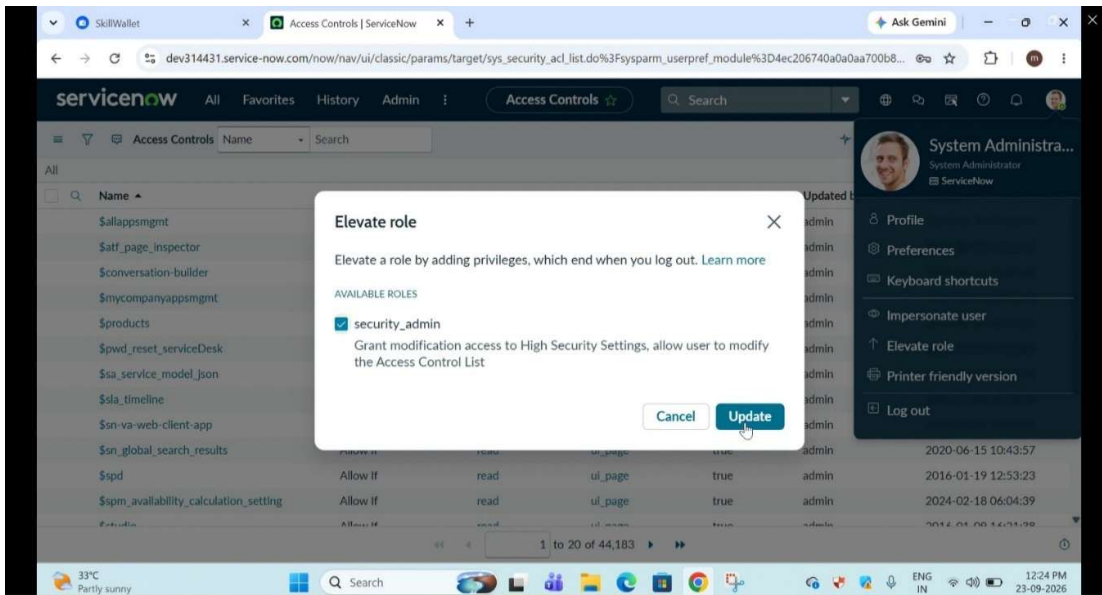
Milestone-3:

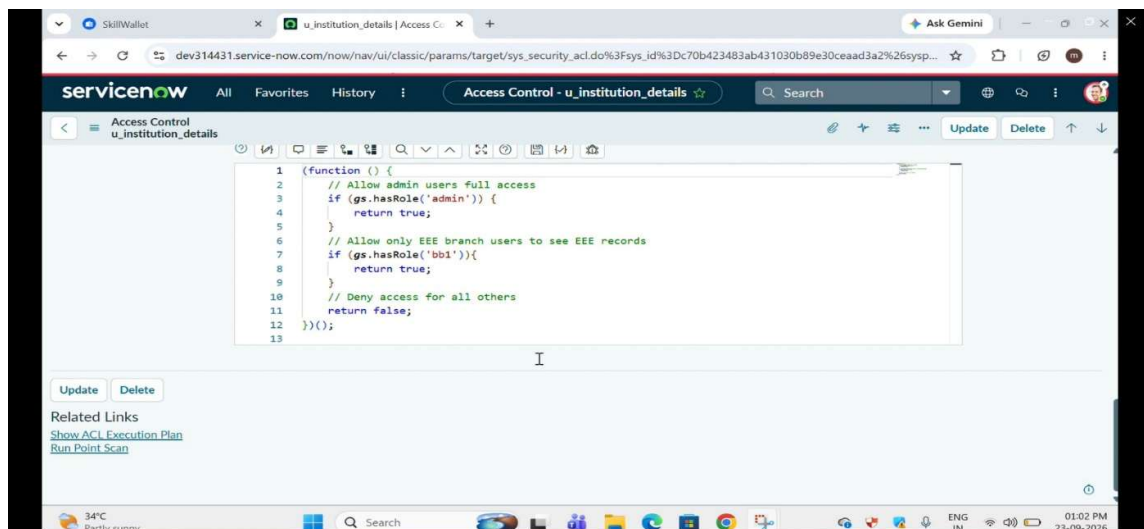
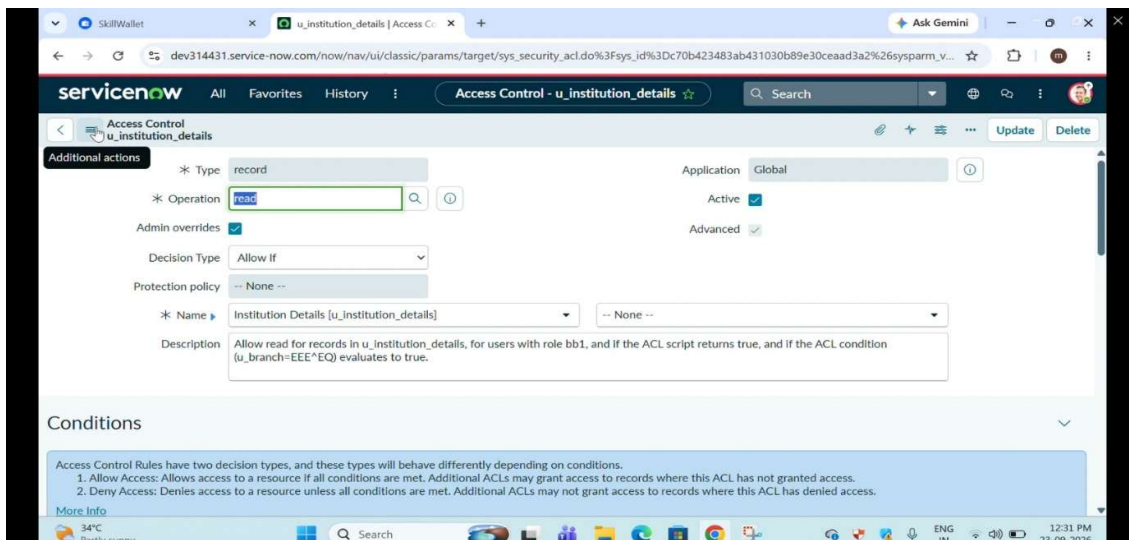
Creation of Access Control Lists(Read ACL):

1. In the **Application Navigator**, search for **System Security** → **Access Control (ACL)**.
2. Elevate "Security_admin" role

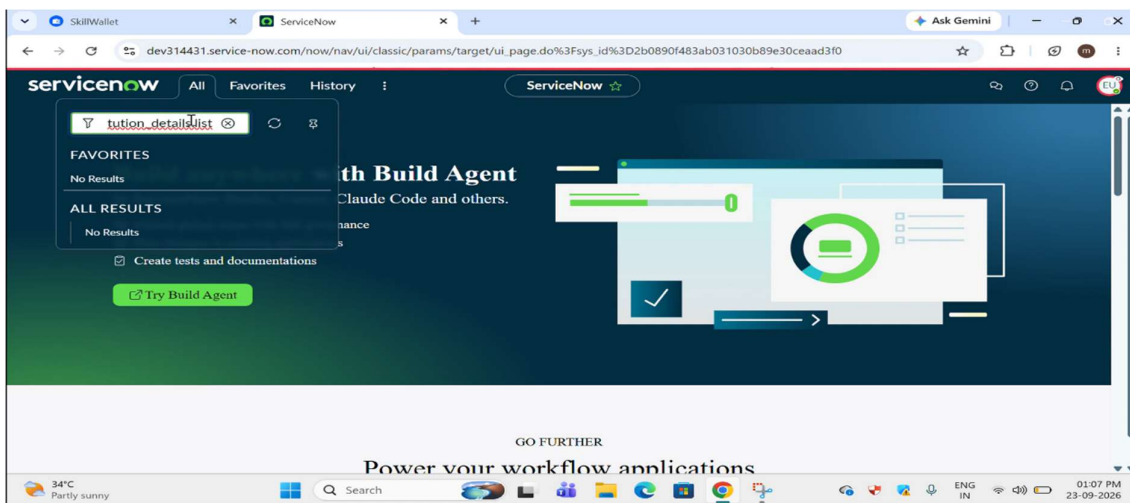
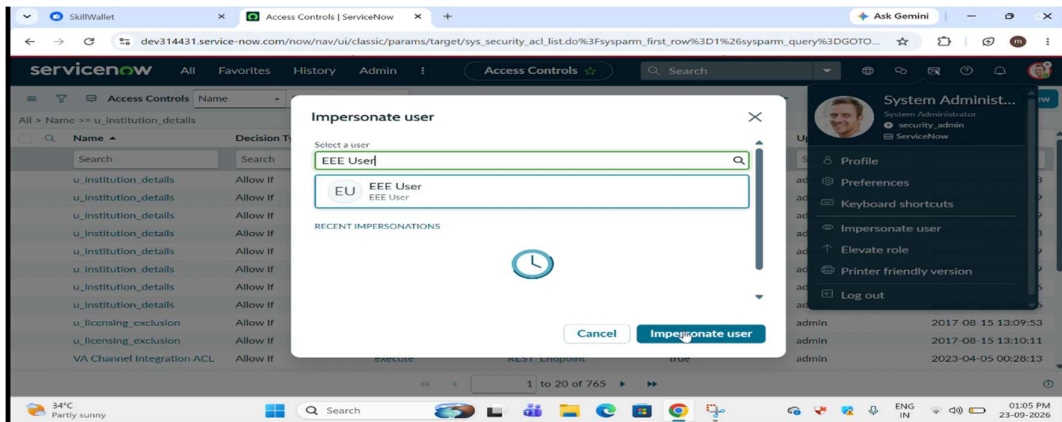
3. Click **New** to create a new ACL.
4. Enter the following details:
 - **Type:** record
 - **Operation:** read
 - **Name:** u_institution_details
 - **Active:** true
 - **Advanced:** true
5. In the **Requires role** related list, add:
 - bb1 (custom role)
6. Add Data condition as **Branch is EEE**
7. In the **Script** field, add the following script:

```
(function () {  
    if (gs.hasRole('admin')) {  
        return true;  
    }  
    if (gs.hasRole('bb1')){  
        return true;  
    }  
    return false;  
})();
```
8. Click **Submit** to save the ACL.



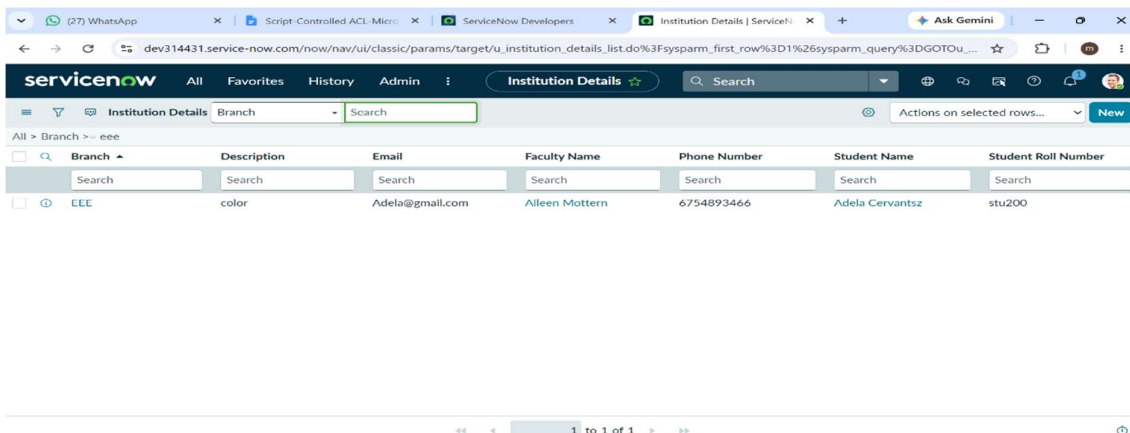


9. Impersonate **EEE users** and open the Student Records list.
10. Copy the table name `u_institution_details.list/LIST` and paste in the All menu.
11. Impersonate **Admin** and open the Student Records list.

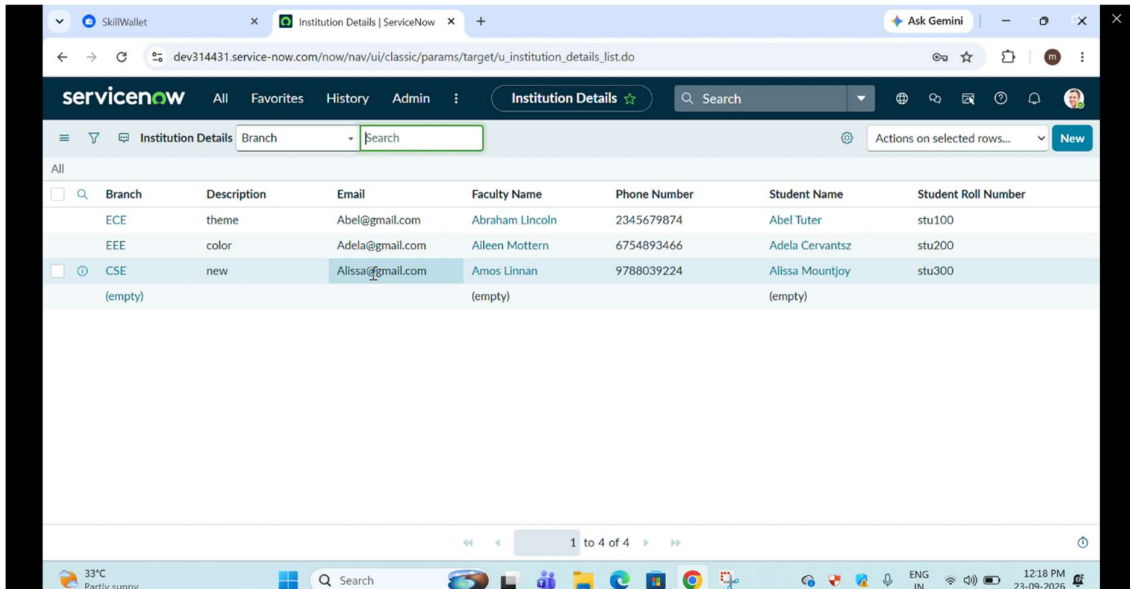


Verify :

- User with **bb1** role can view only **EEE** branch records



- User without the role cannot view any records
- Admin user can view **all records regardless of branch**



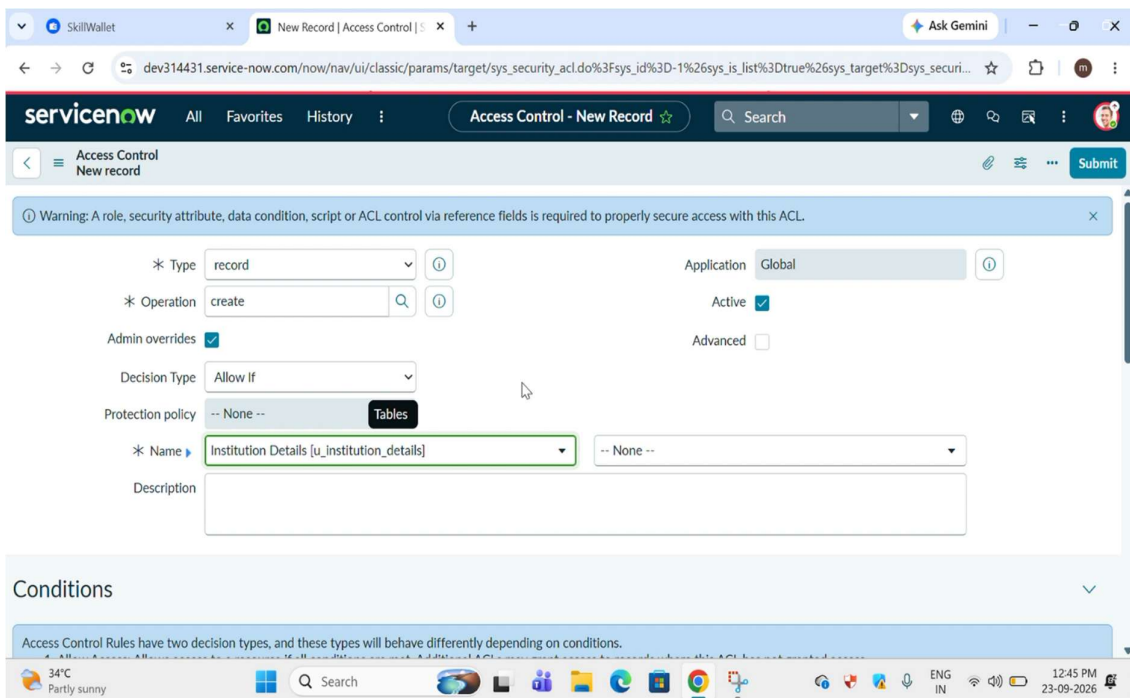
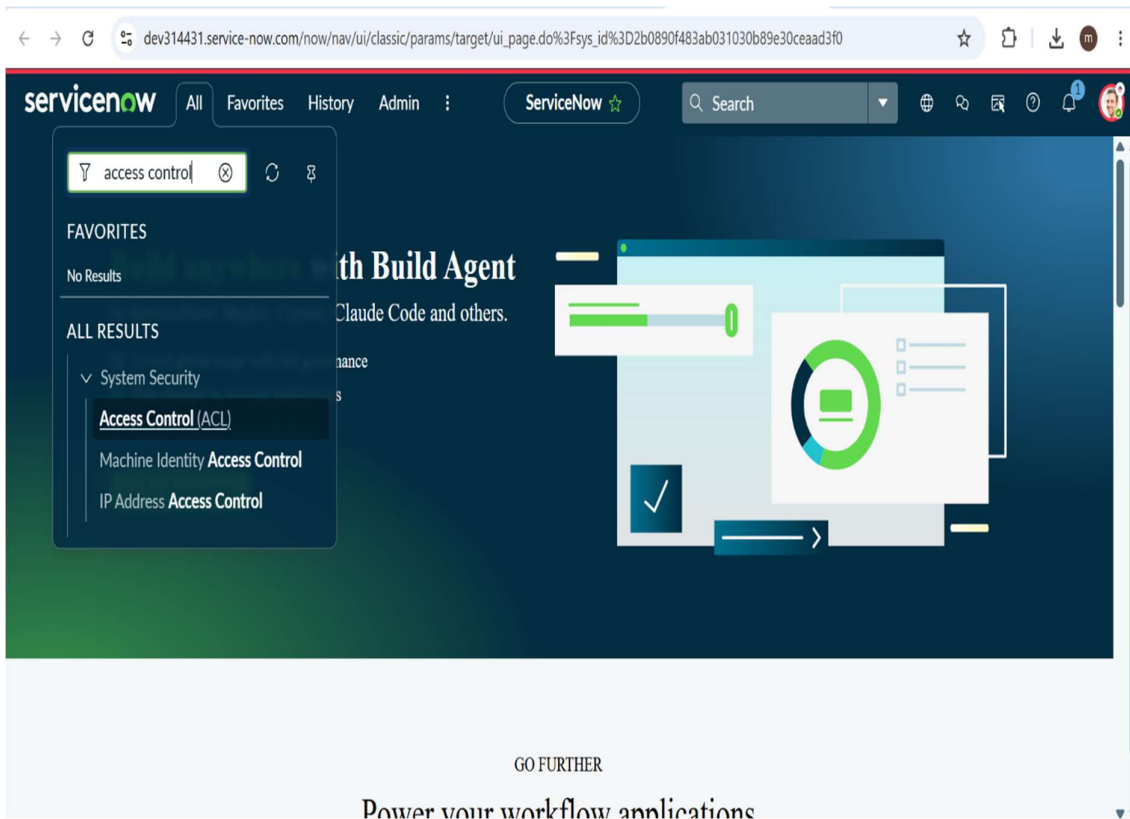
The screenshot shows a ServiceNow interface for 'Institution Details'. The table has columns: Branch, Description, Email, Faculty Name, Phone Number, Student Name, and Student Roll Number. The third record is highlighted.

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
ECE	theme	Abel@gmail.com	Abraham Lincoln	2345679874	Abel Tuter	stu100
EEE	color	Adela@gmail.com	Aileen Mottern	6754893466	Adela Cervantsz	stu200
CSE	new	Alissa@gmail.com	Amos Linnan	9788039224	Alissa Mountjoy	stu300
(empty)			(empty)		(empty)	

Milestone-3:

Creation of Access Control Lists(Create ACL)

1. Click **New** to create a new ACL(Create)
2. Enter the following details:
 - **Type:** record
 - **Operation:** create
 - **Name:** u_institution_details
 - **Active:** true
3. In the **Requires role** related list, add:
 - bb2 (custom role)
4. No need to add any Data condition
5. Save the ACL Record



Access Control - New Record

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role
bb2

[Insert a new row...](#)

Security Attribute Condition

Local or Existing ☒ Existing ☐ Local

Security Attribute

Condition All of these conditions must be met

Verify :

- User with bb1 & bb2 roles can view **only** **EEE** branch records **b** New Button

Institution Details

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE	color	Adela@gmail.com	Aileen Mottern	6754893466	Adela Cervantsz	stu200

Milestone-5:

Creation of Access Control Lists(Write ACL)

1.Click New to create a new ACL(Write)

2. Enter the following details:

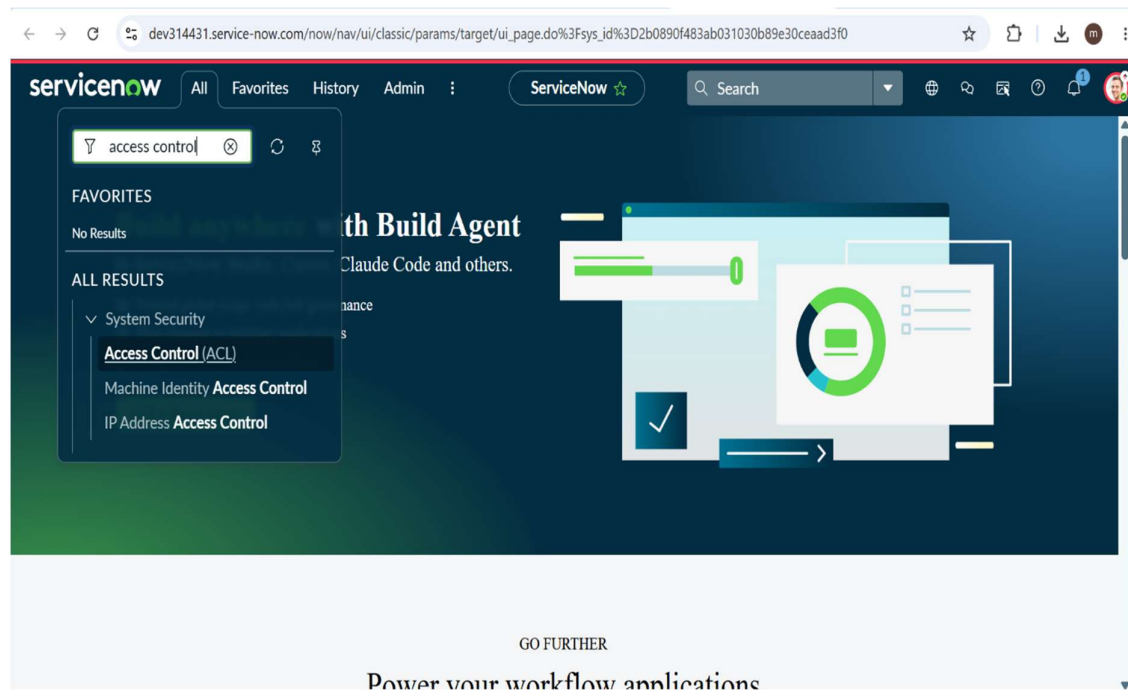
- Type: record
- Operation: Write
- Name: u_institution_details
- Active: true

3. In the Requires role related list, add:

- bb3 (custom role)

4. No need to add any Data condition

5. Save the ACL Record



dev314431.service-now.com/now/nav/ui/classic/params/target/sys_security_acl_list.do

servicenow All Favorites History Admin Access Controls Search

Access Controls Name Search Actions on selected rows... New

All

Name	Decision Type	Operation	Type	Active	Updated by	Updated
\$allappsmgmt	Allow If	read	ui_page	true	admin	2019-02-20 01:02:07
\$att_page_inspector	Allow If	read	ui_page	true	admin	2020-03-20 15:16:15
\$conversation-builder	Allow If	read	ui_page	true	admin	2019-06-14 15:13:38
\$mycompanyappsmgmt	Allow If	read	ui_page	true	admin	2019-02-21 02:31:44
\$products	Allow If	read	ui_page	true	admin	2020-03-10 03:19:22
\$pwd_reset_serviceDesk	Allow If	read	ui_page	true	admin	2023-05-29 05:26:27
\$sa_service_model_json	Allow If	read	ui_page	true	admin	2015-10-28 02:03:33
\$sla_timeline	Allow If	read	ui_page	true	admin	2016-11-14 01:14:39
\$sn-vn-web-client-app	Allow If	read	ui_page	true	admin	2018-03-06 10:20:09
\$sn_global_search_results	Allow If	read	ui_page	true	admin	2020-06-15 10:43:57
\$spd	Allow If	read	ui_page	true	admin	2016-01-19 12:53:23
\$spm_availability_calculation_setting	Allow If	read	ui_page	true	admin	2024-02-18 06:04:39

1 to 20 of 44,184

34°C Partly sunny 12:45 PM 23-09-2026

dev314431.service-now.com/now/nav/ui/classic/params/target/sys_security_acl.do%3Fsys_id%3D1%26sys_is_list%3Dtrue%26sys_target%3Dsys_securi...

servicenow All Favorites History Access Control - New Record Search

Access Control New record Submit

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type record Application Global

* Operation write Active ☒

Admin overrides ☒ Advanced ☐

Decision Type Allow If

Protection policy -- None -- Tables

* Name Institution Details [u_institution_details] -- None --

Description

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

34°C Partly sunny 12:49 PM 23-09-2026

Access Control - New Record

Description

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role: **Save (Enter)**

Security Attribute Condition

Local or Existing: ☒ Existing ☐ Local

Verify :

- User with **bb1 &bb2 &bb3** roles can view and edit the **EEE** branch records along with **New Button**

ServiceNow Institution Details

Branch: **Search**

Actions on selected rows... **New**

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE	color	Adela@gmail.com	Aileen Mottern	6754893466	Adela Cervantsz	stu200

1 to 1 of 1

Milestone-6:

Creation of Access Control Lists(Delete ACL)

1.Click **New** to create a new ACL(Delete)

2.Enter the following details:

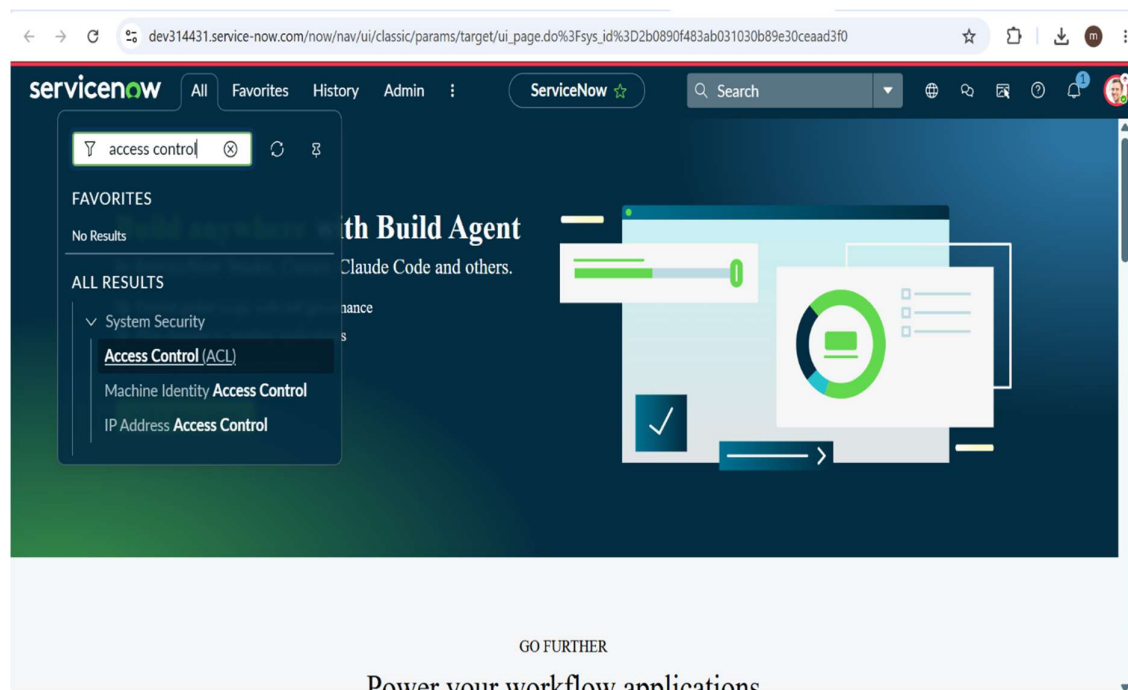
- **Type:** record
- **Operation:** Delete
- **Name:** u_institution_details
- **Active:** true

3.In the **Requires role** related list, add:

- bb4 (custom role)

4.No need to add any Data condition

5.Save the ACL Record



dev314431.service-now.com/now/nav/ui/classic/params/target/sys_security_acl_list.do

skillwallet

Access Controls | ServiceNow

Ask Gemini

serviceNow

All Favorites History Admin

Access Controls

Search

Actions on selected rows...

New

All

Name	Decision Type	Operation	Type	Active	Updated by	Updated
\$allappsmgmt	Allow If	read	ui_page	true	admin	2019-02-20 01:02:07
\$atf_page_inspector	Allow If	read	ui_page	true	admin	2020-03-20 15:16:15
\$conversation-builder	Allow If	read	ui_page	true	admin	2019-06-14 15:13:38
\$mycompanyappsmgmt	Allow If	read	ui_page	true	admin	2019-02-21 02:31:44
\$products	Allow If	read	ui_page	true	admin	2020-03-10 03:19:22
\$pwd_reset_serviceDesk	Allow If	read	ui_page	true	admin	2023-05-29 05:26:27
\$sa_service_model_json	Allow If	read	ui_page	true	admin	2015-10-28 02:03:33
\$sla_timeline	Allow If	read	ui_page	true	admin	2016-11-14 01:14:39
\$sn-vn-web-client-app	Allow If	read	ui_page	true	admin	2018-03-06 10:20:09
\$sn_global_search_results	Allow If	read	ui_page	true	admin	2020-06-15 10:43:57
\$spd	Allow If	read	ui_page	true	admin	2016-01-19 12:53:23
\$spm_availability_calculation_setting	Allow If	read	ui_page	true	admin	2024-02-18 06:04:39

1 to 20 of 44,186

34°C Partly sunny

Search

ENG IN

12:52 PM 23-09-2026

dev314431.service-now.com/now/nav/ui/classic/params/target/sys_security_acl.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dsys_securi...

skillwallet

New Record | Access Control

Ask Gemini

serviceNow

All Favorites History

Access Control - New Record

Search

Submit

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record

* Operation: delete

Admin overrides: ☒

Decision Type: Allow If

Protection policy: -- None --

* Name: Institution Details [u_institution_details]

Description:

Application: Global

Active: ☒

Advanced: ☐

Conditions

of Rules have two decision types, and these types will behave differently depending on conditions.

34°C Partly sunny

Search

ENG IN

12:53 PM 23-09-2026

dev314431.service-now.com/now/nav/ui/classic/params/target/sys_security_acl.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dsys_securi...

servicenow All Favorites History Access Control - New Record Search

Access Control New record Submit

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role Save (Enter)

Security Attribute Condition

Local or Existing ☒ Existing ☐ Local

Security Attribute

Condition All of these conditions must be met

-- choose field --

34°C Partly sunny Search ENG IN 12:53 PM 23-09-2026

Verify :

- Users with **bb1 &bb2 &bb3&bb4** roles can see, create and edit along with delete the **EEE** branch records.

dev314431.service-now.com/now/nav/ui/classic/params/target/u_institution_details_list.do%3Fsysparm_first_row%3D1%26sysparm_query%3D%3DGOTOu...

servicenow All Favorites History Admin Institution Details Search

Institution Details Branch Search Actions on selected rows... New

All > Branch >> eee

☐	Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
☐	Search	Search	Search	Search	Search	Search	Search
☐	EEE	color	Adela@gmail.com	Aileen Mottern	6754893466	Adela Cervantsz	stu200

1 to 1 of 1

Outcome :

Milestone-7

By completing this lab, learners gain a comprehensive understanding of how script-controlled READ, WRITE, CREATE, and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation, and deletion across forms, lists, and Service Portal interfaces.